

Exercise 2: Implementing the Factory Method Pattern

CODE:

```
interface Document {
    void open();
}
class WordDocument implements Document {
    public void open() {
        System.out.println("Opening Word Document...");
    }
}
class PdfDocument implements Document {
    public void open() {
        System.out.println("Opening PDF Document...");
    }
}
class ExcelDocument implements Document {
    public void open() {
        System.out.println("Opening Excel Document...");
    }
}
abstract class DocumentFactory {
    public abstract Document createDocument();
}
class WordDocumentFactory extends DocumentFactory {
    public Document createDocument() {
        return new WordDocument();
    }
}
class PdfDocumentFactory extends DocumentFactory {
    public Document createDocument() {
        return new PdfDocument();
    }
}
class ExcelDocumentFactory extends DocumentFactory {
    public Document createDocument() {
        return new ExcelDocument();
    }
}
public class Main {
    public static void main(String[] args) {
        DocumentFactory wordFactory = new WordDocumentFactory();
```

```
Document wordDoc = wordFactory.createDocument();
wordDoc.open();
DocumentFactory pdfFactory = new PdfDocumentFactory();
Document pdfDoc = pdfFactory.createDocument();
pdfDoc.open();
DocumentFactory excelFactory = new ExcelDocumentFactory();
Document excelDoc = excelFactory.createDocument();
excelDoc.open();
    }
}
```

OUT PUT :

